

VADUZ, Liechtenstein

WMO: 069900

Lat: 47.1275N

Lon: 9.5175E

Elev: 463

StdP: 95.89

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.3	(c) -7.1	(d) -13.2	(e) 1.3	(f) -5.3	(g) -11.1	(h) 1.5	(i) -4.7	(j) 11.3	(k) 12.4	(l) 9.1	(m) 11.6	(n) 1.3	(o) 330	(p) 0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 30.2	(d) 20.0	(e) 28.4	(f) 19.3	(g) 26.8	(h) 18.6	(i) 21.1	(j) 28.2	(k) 20.2	(l) 26.8	(m) 19.3	(n) 25.4	(o) 3.5	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.2	24.6	17.7	13.4	23.4	17.0	12.8	22.5	63.1	28.3	59.8	27.0	56.8	25.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.4	7.0	5.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-11.6	33.0	2.7	1.7	-13.5	34.2	-15.1	35.2	-16.6	36.2	-18.6	37.4
			-11.9	23.0	2.7	1.0	-13.9	23.8	-15.5	24.3	-17.0	24.9	-19.0	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	1.5	3.1	7.2	10.9	15.0	18.3	19.9	19.5	15.4	11.6	6.1	2.3		
	DBStd	7.76	4.86	4.88	4.45	4.46	3.94	4.12	3.49	3.48	3.54	3.99	4.66	4.54		
	HDD10.0	1026	264	197	111	43	5	1	0	0	2	29	134	241		
	HDD18.3	2915	521	427	347	225	117	52	24	27	102	210	368	496		
	CDD10.0	1370	1	3	22	69	161	249	306	296	163	79	17	3		
	CDD18.3	219	0	0	0	2	14	51	72	65	12	2	1	0		
	CDH23.3	1499	0	0	0	11	109	389	519	404	57	9	0	0		
	CDH26.7	385	0	0	0	1	17	106	149	107	4	1	0	0		
(14) Wind	WSAvg	2.0	2.1	2.3	2.3	2.5	2.1	1.9	1.8	1.7	1.6	1.8	2.0	2.1		
(15) Precipitation	PrecAvg	1270	73	77	85	86	116	143	152	161	112	86	95	86		
	PrecMax	1720	172	244	192	158	256	236	271	290	178	202	243	220		
	PrecMin	973	5	24	29	23	22	60	69	36	46	6	1	7		
	PrecStd	156	45	51	43	36	54	49	52	55	38	44	50	48		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	16.7	20.5	24.8	28.9	32.1	32.7	32.1	27.1	24.5	20.6	17.8		
		MCWB	7.9	7.9	10.7	13.7	17.1	20.3	20.7	20.7	17.6	14.1	11.4	9.3		
	2%	DB	12.9	14.2	18.1	22.7	26.2	29.5	30.2	29.4	25.1	22.0	18.7	14.4		
		MCWB	6.1	6.6	9.8	12.8	16.1	19.4	20.0	20.4	17.5	13.6	10.2	7.1		
	5%	DB	10.5	12.2	16.0	20.6	24.1	27.5	28.2	27.4	23.1	19.8	16.2	10.6		
		MCWB	5.3	5.8	8.9	11.8	15.0	18.8	19.6	19.6	16.4	13.1	9.4	6.0		
	10%	DB	8.1	9.8	14.1	18.5	22.1	25.5	26.3	25.5	21.1	17.7	12.6	8.3		
		MCWB	4.6	5.1	7.9	10.8	14.4	17.9	18.8	18.8	15.8	12.4	8.4	5.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.5	8.7	11.9	14.9	18.8	21.8	22.3	22.2	19.7	16.2	12.0	9.4		
		MCDB	13.1	14.4	18.8	23.3	26.5	29.9	30.3	29.5	25.1	20.3	18.3	16.6		
	2%	WB	7.0	7.5	10.4	13.5	17.1	20.4	21.0	21.1	18.3	15.0	10.8	8.0		
		MCDB	11.4	12.8	16.4	21.1	23.9	27.7	28.3	27.8	23.2	18.9	16.3	12.2		
	5%	WB	5.9	6.4	9.4	12.3	15.9	19.3	20.0	20.1	17.2	14.0	9.9	6.8		
		MCDB	9.7	11.1	14.9	19.2	22.3	26.1	26.8	26.3	21.5	18.3	15.1	9.9		
	10%	WB	4.7	5.3	8.4	11.2	14.8	18.3	19.1	19.1	16.2	13.1	8.9	5.5		
		MCDB	7.7	9.3	13.2	17.4	20.6	24.4	25.2	24.6	20.0	16.6	12.4	8.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.9	8.6	9.4	9.4	9.3	9.4	8.7	8.2	7.9	6.5	5.8		
		MCDBR	9.6	10.1	11.2	11.8	12.2	12.2	12.1	11.2	10.3	10.4	10.4	9.3		
		MCWBR	5.1	5.4	6.0	5.9	5.8	5.8	5.6	5.6	5.4	5.1	4.7	4.8		
		MCDBR	8.1	9.0	10.5	11.4	11.2	11.2	11.2	10.4	9.4	9.2	8.9	7.7		
	5% WB	MCWBR	4.6	5.3	5.8	6.1	5.7	5.6	5.5	5.6	5.3	5.0	4.6	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.279	0.307	0.351	0.398	0.415	0.423	0.424	0.415	0.380	0.347	0.311	0.277		
	taud		2.350	2.276	2.234	2.213	2.237	2.265	2.265	2.308	2.381	2.450	2.466	2.398		
	Ebn at Noon		811	855	863	854	851	844	837	829	827	803	769	771		
	Edh at Noon		72	96	117	132	134	131	130	119	101	81	65	61		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.20	3.41	4.38	4.91	5.68	5.56	4.84	3.68	2.36	1.37	1.09		
	RadStd		0.15	0.28	0.47	0.66	0.49	0.52	0.50	0.48	0.37	0.27	0.18	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.97	+0.40	N/A	N/A	N/A	+100	+48
(47) Regional (120 neighbors)		N/A	N/A	N/A	+0.68	+0.41	+0.33	N/A	N/A	+57	+15

Nomenclature: See separate page